Month 1: Purple Teaming Batch 01

# Network Security & Protocol Analysis

## Week 2: Network Discovery & Reconnaissance

Instructor: **Muhammad Saad**

# Passive Reconnaissance

**Passive reconnaissance** means **collecting information about a target without directly interacting with the target system**.

The target **does not know** that someone is gathering information.

- Searching information on **Google**
- Checking **social media profiles**

# Active Reconnaissance

**Active reconnaissance** means **directly interacting with the target system to gather information**.

Because the attacker communicates with the target, **the target may detect the activity**.

- Scanning a website or server
- Sending packets to check open ports

# ARP - Address Resolution Protocol

Address Resolution Protocol (ARP) is a protocol used to map an **IP address** to a **MAC address** inside a local network (LAN).

**Why ARP is Needed?**

Devices communicate over:

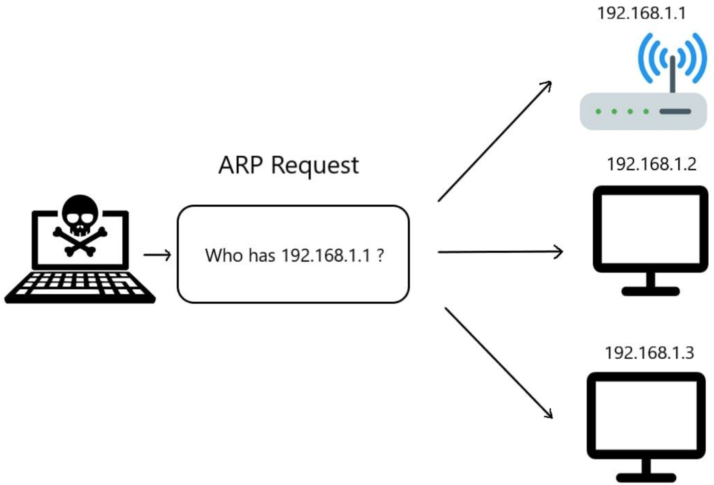
- **IP address** → Logical address (Layer 3)
- **MAC address** → Physical hardware address (Layer 2)

When a device wants to send data to another device on the same network, it must know the destination **MAC address**. ARP helps find It.

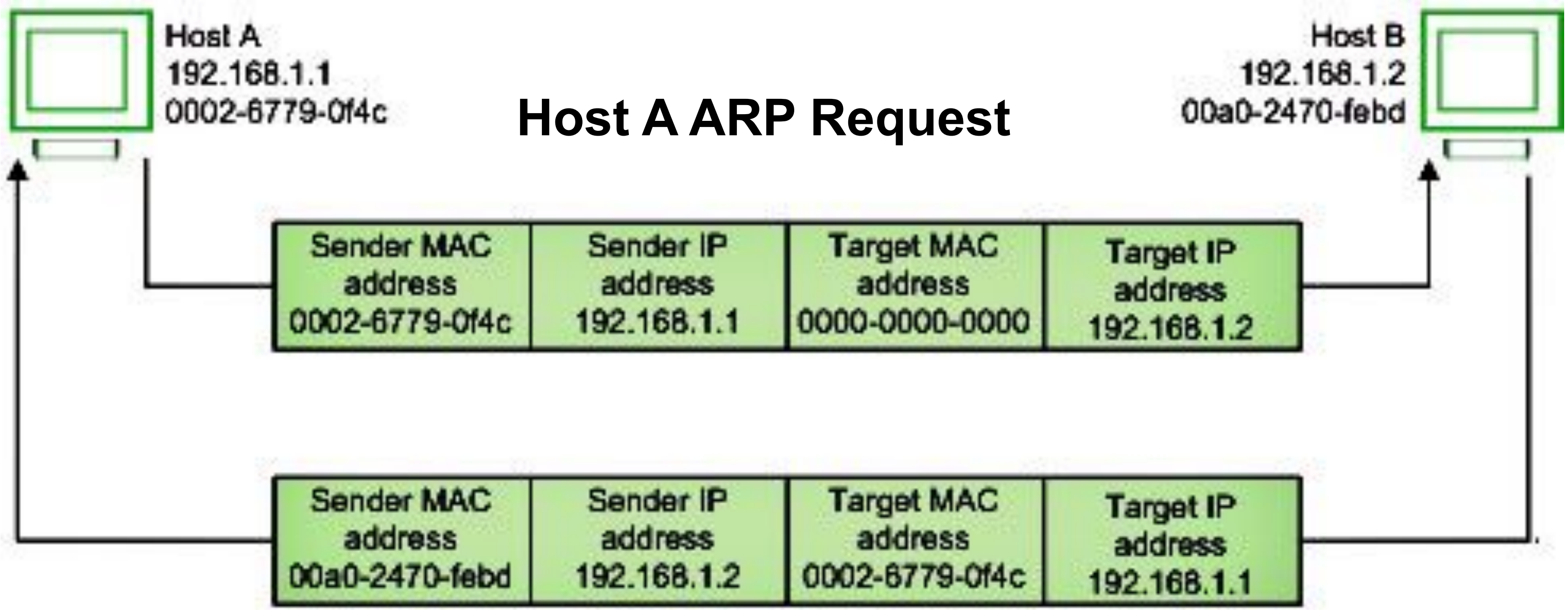
**How ARP Works ?**

- A device sends an **ARP Request** (broadcast): "Who has 192.168.1.1?"
The device with that IP replies with its **MAC address**.

**Key Point:** Works only within a local network & operates at Layer 2 (Data Link Layer).



# Host A ARP Request




# Host B Response

# ICMP - Internet Correct Message Protocol

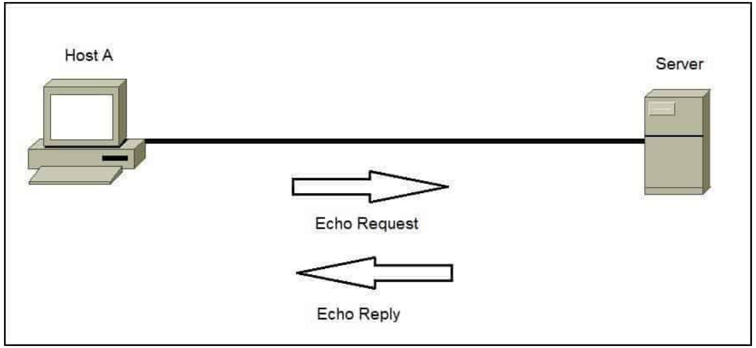
Internet Control Message Protocol (ICMP) is a protocol used for **network diagnostics and error reporting**.

It helps devices understand:

- **Whether another device is reachable**
- **If a route is unavailable**
- **If packets are being dropped**

**Common ICMP Uses Ping**

- Sends ICMP Echo Request: An **Echo Request** is a message sent by one device to another asking: **“Are you reachable?”**
- Receives ICMP Echo Reply: An **Echo Reply** is the response sent back to the sender saying: **“Yes, I am reachable.”**




**Key Point:** Works at **Layer 3 (Network Layer)**, used for **testing and troubleshooting connectivity & can operate across different networks (not limited to LAN)**

Q1) ICMP & ARP based attacks do belong to what fundamental method used to gather information about a target ? (Passive Reconnaissance/Active Reconnaissance)

Q2) Which technique of Active Recossainace is used to gather target's information within the same network ? (ICMP/ARP)

# ICMP-Based Active Discovery

### ICMP Echo Request (Ping Sweep):

- A host sends an **ICMP Echo Request**
- Active devices reply with an **ICMP Echo Reply**
- Responding IPs are marked as "alive”

### Example Tools

- `ping`
- `fping`
- `hping`
- `nmap -sn`

# ARP-Based Active Discovery

### ARP discovery sends:

- **ARP Request**: "Who has 192.168.1.10?"
- The target replies with its **MAC address**
- Any reply indicates a live host

### Example Tools

- `arp-scan`
- `netdiscover`
- `nmap -PR`

# TCP SYN (Stealth Scan)

A **TCP SYN Scan** is a fast and commonly used port scanning technique that determines whether a TCP port is open **without completing the full TCP handshake**.

It is often called a **Stealth Scan** because it is less likely to be logged compared to a full connection scan.

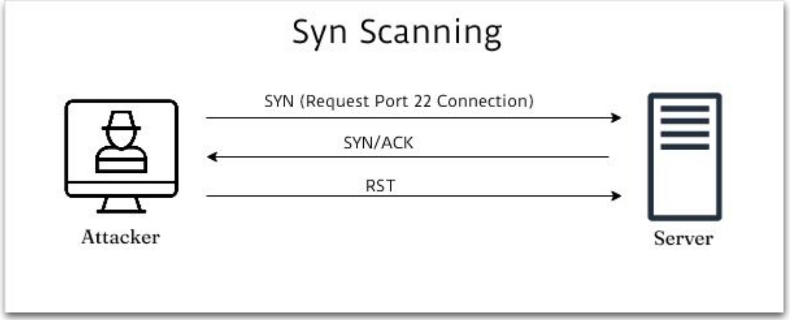
**How SYN Scan Works**

Instead of completing the handshake, the scanner:

1. Sends a **SYN** packet to the target port.
2. Waits for the response.
3. Does **not** send the final ACK.

**If the port is open:** Scanner sends **RST** to terminate the connection

immediately & the handshake is never completed.




| | |
|---|---|
| SYN | Sender request for communication |
| SYN-ACK | Receiver respond means port is open |
| RST | Connection closed by sender |

# Best of Luck for the Week 2 Tasks

Instructor: **Muhammad Saad**